

EECE 571Q

Lecture 10: Beneath the surface (part 2)

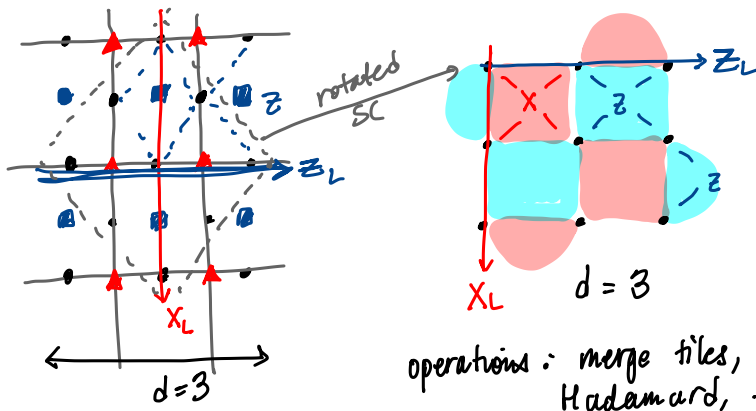
Thursday 11 June 2026

Announcements

- Project proposal due tomorrow Friday 12 June 23:59
 - please advise of any lecture date constraints
 - I will try to order lectures based on content
- Exam reference sheet info available (due Thursday 18 June)
- Class on Zoom next week
- Quiz 5 on Tuesday

Last time

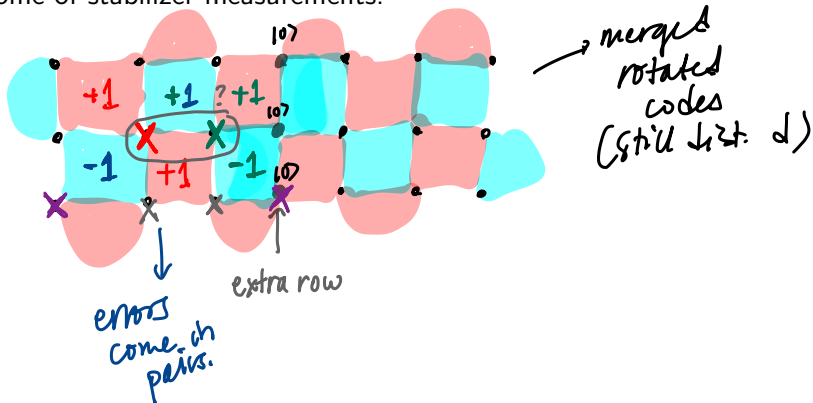
We introduced the **surface code** (2D planar code), and saw some examples of operations that can be performed on it. *(not periodic)*



operations: merge tiles, split, Hadamard, ...

Last time

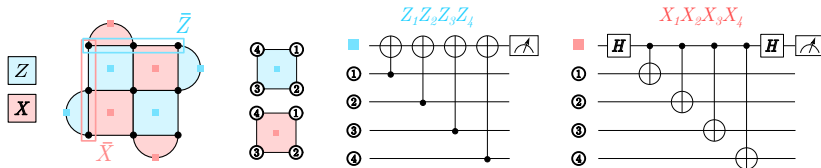
We briefly discussed code distance and how errors manifest in the syndrome of stabilizer measurements.



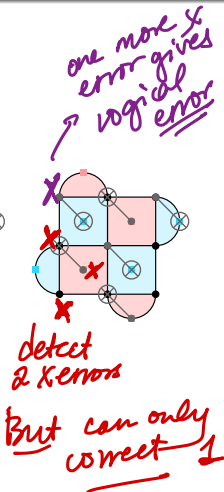
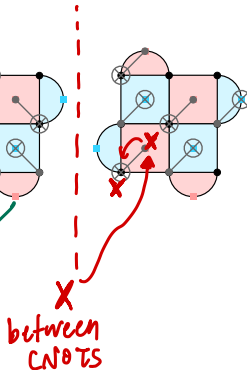
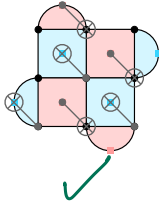
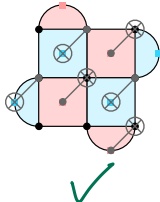
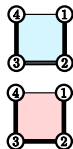
Learning outcomes

- 1 Construct circuits for surface code syndrome measurement
- 2 Describe general decoding strategies, and specific decoding strategies for surface codes
- 3 Carry out simulations in stim to estimate the surface code (pseudo)threshold

Syndrome measurement



Syndrome measurement



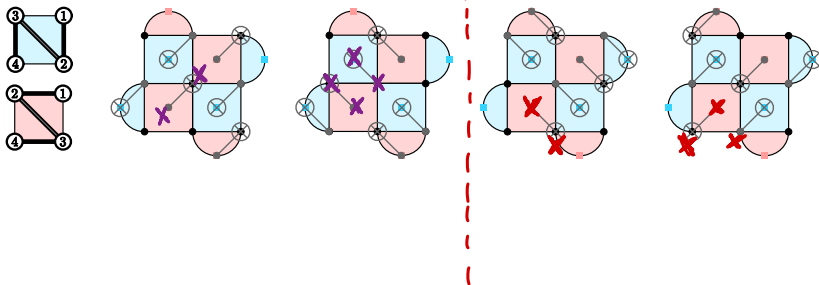
Hook error

one more $\times$
error gives
logical error

detect
2 X errors

But can only
correct 1

Syndrome measurement



Syndromes and cosets

When we studied the 9-qubit code (L4), we found multiple errors that produced the same syndrome.

Syndromes and decoding

Exercise: What are the error syndromes for Y_1 and $X_0Z_1X_2$?

Name	Operator
g_1	$ZZIIIIII$
g_2	$IZZIIIIII$
g_3	$IIIZZIIII$
g_4	$IIIIZZIIII$
g_5	$IIIIIIZZI$
g_6	$IIIIIIIZZ$
g_7	$XXXXXXII$
g_8	$IIIIXXXXX$
Z	$XXXXXXX$
X	$ZZZZZZZ$

Screenshot: Nielsen & Chuang, chapter 10.5.6.

Syndromes and cosets

Why this happens: cosets of $N(S)$ w.r.t. the Pauli group.

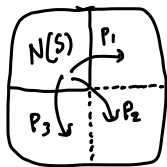
every Pauli that commutes with stabilizer group

Gottesman Proposition 3.8 (rephrased)

Proposition: Two errors, E and F , are in the same coset of $N(S)$ if they have the same error syndrome.

$$F = E \cdot N, N \in N(S)$$

Example: $S = \{ZZI, IZZ\}$.



$$N(S) = \{ \text{I, } ZZI, IZZ, ZIZ, ZZZ, IIZ, ZII, IZI, XXX, YYY, XYY, YXX, YXY, XXY, YXX, XYY \}$$

logical Z → ZZI, IIZ, ZII, IZI
logical X → $XXX, YYY, XYY, YXX, YXY, XXY$

$$|N(S)| = 4 \cdot 2^{n+k}$$

Syndromes and cosets

Similarly: consider cosets of S in $N(S)$.

Gottesman Proposition 3.10 (rephrased)

Proposition: Let S be a stabilizer, and consider $N_1, N_2 \in N(S)$. N_1 and N_2 are in the same coset of S iff, for all codewords $|\psi\rangle$, $N_1|\psi\rangle = N_2|\psi\rangle$.

Maximum likelihood decoding

Decoding is the process of interpreting the syndrome measurement and making a decision about which error actually occurred.

Initial idea: associate each error syndrome with the **lowest-weight representative** of each coset of $N(S)$.

Better option: maximum likelihood decoding.

Proposition 3.13. *Let C_s be the coset of $\hat{N}(S)$ with error syndrome s . $\hat{Q}_s\hat{S}$ is the coset of $\hat{S}$ containing the canonical error $\hat{Q}_s$ with syndrome s , so $\hat{Q}_s\hat{S} \subseteq C_s$.*

a) *The probability of error syndrome s (regardless of whether we correctly identify the error) is*

$$p_s = \sum_{\hat{P} \in C_s} p_P. \quad (3.22)$$

b) *The probability of having syndrome s and no logical error (i.e., error correction is successful) is*

$$p_{s,OK} = \sum_{\hat{P} \in \hat{Q}_s\hat{S}} p_P. \quad (3.23)$$

c) *The total probability of successfully decoding the state is*

$$p_{OK} = \sum_s \sum_{\hat{P} \in \hat{Q}_s\hat{S}} p_P. \quad (3.24)$$

Gottesman Proposition 3.13

Maximum likelihood decoding

Example: bit flip code, measure syndrome 10. X, Y, Z error occurs independently on each qubit with probability $p/3$.

Cosets of $N(S)$:

$(III)00 : \{III, ZZI, IZZ, ZIZ, ZZZ, ZII, IZI, IIZ, XXX, YYY, YXY, XYX, YXX, XYY, XXY, YYX\}$

$(IIX)01 : \{IIX, ZZX, IZY, ZIY, ZZY, ZIX, IZX, ILY, XXI, YYZ, YXZ, XYI, YXI, XYZ, XXZ, YYI\}$

$(XII)10 : \{XII, YZI, XZZ, YIZ, YZZ, YII, XZI, XIZ, IXX, ZYY, ZXY, IYX, ZXX, IYY, IXY, ZYX\}$

$(IXI)11 : \{IXI, ZYI, IYZ, ZXZ, ZYZ, ZXI, IYI, IXZ, XIX, YZY, YIY, XZX, YIX, XZY, XIY, YZX\}$

Probability of syndromes 01, 10, 11: $\frac{2}{3}p(1-p)^2 + \frac{8}{9}p^2(1-p) + \frac{6}{27}p^3$

Probability of syndrome 00: $(1-p)^3 + p(1-p)^2 + \frac{1}{3}p^2(1-p) + \frac{1}{3}p^3$

Maximum likelihood decoding

Good choice: coset representative XII :

$(XII)_{10} : \{XII, YZI, XZZ, YIZ, YZZ, YII, XZI, XIZ, IXX, ZYY, ZXY, IYX, ZXX, IYY, IXY, ZYX\}$

Probability of syndrome 10 and no logical error:

$$\frac{p}{3}(1-p)^2 + \frac{2}{9}p^2(1-p) + \frac{p^3}{27}$$

For *each syndrome*, choose coset representative that *maximizes* probability of no logical error if applied as correction.

Bad choice: ZXX . Coset of S is $\{XZZ, IYX, ZYY, IXY\}$.

Minimum weight perfect matching

We must also consider how measurement errors propagate in time. Perform d rounds of syndrome measurements to identify them.

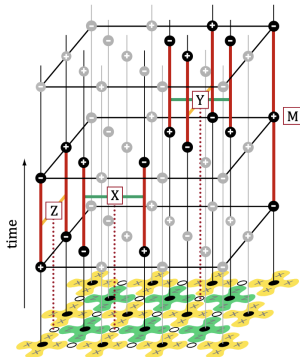


Image: Fowler et al., (2012), *Surface codes: Towards practical large-scale quantum computation*, PRA86 (3) 032324.

Stim simulations

Let's use stim to perform some experiments with various error models and estimate the threshold of the surface code.

Next time

Next class:

- Quiz 5
- Magic states: injection (gate teleportation) and distillation

Action items:

- ① Work on and submit project proposal
- ② Start preparing exam reference sheet

Recommended reading for this / next class:

- Gottesman Ch. 3.4 and 13; N & C 10.6
- Fowler, Mariantoni, Martini, and Cleland (2012), *Surface codes: Towards practical large-scale quantum computation*, Phys. Rev. A 86 (3) 032324.
- Horsman, Fowler, Devitt, and van Meter (2012), *Surface code quantum computing by lattice surgery*, NJP 14 123011.
- [PennyLane tutorial on magic states](#)
- [PennyLane tutorial on state distillation](#)